Cunquan Li (李存权)

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EDUCATION

08/2023 – Present **Ph.D. Student of Materials Science and Engineering**
School of Engineering, Westlake University, Hangzhou, China
Zhejiang University-Westlake University joint Ph.D. program
Advisor: Dr. Naizhou Wang

08/2019 – 06/2023 **Bachelor of Material Physics (B.Sc.)**
School of Materials Science and Engineering
Tianjin University of Technology, Tianjin, China
GPA: 86.7/100 (Rank: 3/112)
Advisor: Prof. Yukai An

RESEARCH EXPERIENCE

01/2025 – Present **Microscopic spectroscopy of 2D (artificial) Moiré superlattices**
Abstract: Coming soon～
As a Ph.D. student; Advisor: Dr. Naizhou Wang

12/2023 – 12/2024 **The Synthesis of 2D halide perovskite heterostructures**
Abstract: By controlling the solution concentration and the growth temperature and time, we have attempted to synthesis highly stable and tunable lateral epitaxial heterostructures, multiheterostructures and superlattices.
As a Ph.D. student; Advisor: Dr. Enzheng Shi

02/2023 – 11/2023 **Synthesis and Application of the novel 2D Janus MoTeX (X=S, Se) monolayer**
Abstract: Based on a $MoTe_2$ monolayer, we have attempted to fully replace the top-layer Te with S or Se atoms by room-temperature atomic-layer substitution. If the Janus structure can be synthesised, it will enrich the materials database and provide more potential for heterostructures.
As a visiting student; Advisor: Dr. Yunfan Guo

09/2020 – 02/2023 **First-Principles Study: Valley polarization and magnetic modulation of novel 2D Janus ferromagnetic monolayers**
Abstract: Combining existing materials science databases with multi-scale calculations (VASP), we predict a large set of potential 2D Janus ferro-valley materials with experimental possibilities.
As an undergraduate student; Advisor: Prof. Yukai An

HONORS & AWARDS

2023 The 18th "Challenge Cup" National College Student Curricular Academic Science and Technology Works Competition; **First Prize**

2023 The 17th "Challenge Cup" Tianjin City College Student Curricular Academic Science and Technology Works Competition; **Grand Prize**

2023 Outstanding Graduates of Tianjin University of Technology

2023 Mingli Scholarship (Grand Award: 1/10)

2022 Tianjin Innovation and Entrepreneurship Scholarship for Undergraduate

2022 Tianjin Government Scholarship for Undergraduate

PUBLICATIONS

Total citations: 87, h-index = 5, i10-index = 4 (Google Scholar Profile, accessed on June 2025)

Peer-reviewed Research Articles († denotes equal contribution | * denotes corresponding authorship)

1. Y. Li, Z. Li, Y. Han, R. Lai, J. Yao, **C. Li**, ..., & E. Shi*. Dual Oxidation Suppression in Lead-Free Perovskites for Low-Threshold and Long-Lifespan Lasing. ***Adv. Mater.***, 37, 2418931. (2025) [Link]
2. Jiashuo Yan, **Cunquan Li**, Huijuan Sun*, and Yukai An*, Intrinsic origin of room temperature ferromagnetic ordering in $(In_{0.97-x}Co_xMg_{0.03})_2O_3$ thin films, ***Vacuum***, 224, 113102. (2024) [Link]
3. **Cunquan Li**†, Jiashuo Yan†, Yankai Chen, and Yukai An*, Structural characterization, electrical and magnetic properties of $(In_{0.97-x}Co_xMg_{0.03})_2O_3$ films, ***Phys. Lett. A***, 494, 129282. (2024) [Link]
4. **Cunquan Li** and Yukai An*, Two-dimensional rare-earth Janus 2*H*−Gd*XY* (*X*, *Y*=Cl, Br, I; *X*≠*Y*) monolayers: Bipolar ferromagnetic semiconductors with high Curie temperature and large valley polarization, ***Phys. Rev. B***, 107, 115428. (2023) [Link]
5. **Cunquan Li** and Yukai An*, Two-dimensional ferromagnetic semiconductors of rare-earth Janus 2*H*-GdIBr monolayers with large valley polarization, ***Nanoscale***, 15, 8304. (2023) [Link]
6. **Cunquan Li** and Yukai An*, Two-dimensional intrinsic ferrovalley Janus 2*H*-VSeS monolayer with high Curie temperature and robust valley polarization, ***Phys. Rev. Mater.***, 6, 094012. (2022) [Link]
7. **Cunquan Li** and Yukai An*, Tunable magnetocrystalline anisotropy and valley polarization in an intrinsic ferromagnetic Janus 2*H*-VTeSe monolayer, ***Phys. Rev. B***, 106, 115417. (2022) [Link]
8. Yimo Hou, **Cunquan Li**, and Yukai An*, Effects of substitutional Fe and oxygen vacancies on room temperature ferromagnetism of Fe-doped In_2O_3 dilute magnetic nanowires, ***Physica E Low Dimens. Syst. Nanostruct.***, 138, 115129. (2022) [Link]

GRANTS

1. National Training Program of Innovation and Entrepreneurship for Undergraduates (Grant No. 202110060001), "Structural characterization, electrical and magnetic properties of $(In_{0.97-x}Fe_xMg_{0.03})_2O_3$ films" , 1/1/2021-12/31/2022
2. National Training Program of Innovation and Entrepreneurship for Undergraduates (Grant No. 201910060001), "Growth of In_2O_3 dilute magnetic nanowires and their local structure and spin regulation" , 1/1/2020-12/31/2021

SERVICE

Peer Review for Scientific Journals: **APS**: *Physical Review & Physical Review Letters*

SKILLS

First-principle calculations: Magnetic properties (Curie/Néel temperature, Magnetocrystal Anisotropy); Electronic properties (ELF/Band/DOS/effective mass); Optical properties (Raman/Absorption); Topological properties (Surface state, TB model with Wannier functions, Berry curvature, Anomalous/Spin Hall effect), etc.

Synthesis and Characterization of Materials：	CVD/PVD/CVT, Janus Conversion, Solvent Method, Wet/Dry Transfer, Mechanical Exfoliation, *f*s/*n*s Laser Etching; Confocal Raman/PL/Lasing, AFM, SEM/EDS, PXRD, etc.
Other skills:	Python; MATLAB; LaTeX; Linux; MS office; OriginPro; Mathematical modelling; Machine learning, etc.